Minda Zhao

mindazhao@gatech.edu | minda-zhao.github.io/Personal-Website
Room 1111, George Tower, 55 5th St NW, Atlanta, GA 30332

Education

Georgia Institute of Technology (Georgia Tech) — Atlanta, GA
Ph.D. candidate in Operations Research | Advisor: Prof. Xin Chen — 2022–2027 (Expected)

University of Illinois, Urbana-Champaign (UIUC) — Champaign, IL
Ph.D. student in Industrial Engineering — 2021–2022
Transferred to Georgia Institute of Technology.

University of Chicago (UChicago) — Chicago, IL
M.S. in Computational and Applied Mathematics | Advisor: Prof. Lek-Heng Lim — 2019–2020

Sun Yat-sen University (SYSU) — Guangzhou, China
B.S. in Information and Computing Science — 2015–2019

Research Interests

My research lies at the intersection of Artificial Intelligence (AI) and Operations Research (OR), spanning theory and practice in AI-driven operations management. I am particularly interested in:

- **Benign nonconvex landscapes:** Uncovering structural properties of nonconvex problems in reinforcement learning and operations management that enable gradient-based methods to converge to globally optimal solutions.
- **Human-AI interaction:** Designing policies to improve human supervision of autonomous AI in operations, supported by analytical modeling and large-scale field experimentation.

Applications include inventory management (such as smart vending replenishment), queueing systems in service operations, and cash management.

Preprints

1. **Benign Nonconvex Landscape for Policy Optimization: Infinite-Horizon Discounted MDPs with General State and Action Spaces.** [ArXiv] [SSRN]
   Xin Chen, **Minda Zhao** (alphabetical order)
   – Under review at *Management Science*.
   – Extended abstract accepted by NeurIPS 2026 ML×OR Workshop.
2. **Hidden Convexity in Queueing Models.** [ArXiv] [SSRN]
   Xin Chen, Linwei Xin, **Minda Zhao** (alphabetical order)
   – Under minor revision at *Operations Research*.
   – Selected to present at Analytics for X Conference, 2026.
3. **A Simple Solution to Improving Human Supervision of Algorithms: Evidence from Smart Vending.** [ArXiv] [SSRN]
   Minda Zhao, Brian Han, Xin Chen, Tao Zhu
   – Under major revision at *Manufacturing & Service Operations Management*.
   – **Finalist**, INFORMS BOM Section Best Working Paper Competition, 2026.
   – **Finalist**, Best Student Paper Award, AI in Business Conference, 2026.
   – **Winner**, POMS College of BOM Junior Scholar Paper Competition, 2026.
   – Selected as an oral presentation at MSOM Supply Chain SIG, 2025.
   – **Third Place**, CSAMSE Practice Award 2024.
4. **Landscape of Policy Optimization for Finite Horizon MDPs with General State and Action.** [ArXiv]
   Xin Chen, Yifan Hu, **Minda Zhao** (alphabetical order)
   – Under minor revision at *Operations Research*.
   – Extended abstract accepted by NeurIPS 2025 ML×OR Workshop.

Publications

1. **Stochastic Steffensen Method.** [Link]
   Minda Zhao, Zehua Lai, Lek-Heng Lim. 2024
   Computational Optimization and Applications, pp. 1–32.

Honors and Awards

Finalist, INFORMS BOM Section Best Working Paper Competition 2026
Finalist, Best Student Paper Award, AI in Business Conference 2026
Finalist, Next-Gen Scholar's Symposium 2026
Winner, POMS College of BOM Junior Scholar Paper Competition 2026
Shabbir Ahmed PhD Fellowship for Excellence in Research 2025
Awarded annually to one Ph.D. student (co-winner) in ISyE for research in optimization
Third Place, CSAMSE Practice Award 2024
NCCR Automation Fellowship, Switzerland (CHF 18,000; approx. US $22,000) 2024
Hansen Fellowship, UIUC 2021

Industry Experience

Shenzhen Fengyi Technology Co., Ltd. Shenzhen, China
Research Scientist Intern May 2023–Aug 2023
Developed a new replenishment strategy, reducing annual inventory costs by $1,000,000.

Cainiao Network Hangzhou, China
Research Scientist Intern Jun 2021–Oct 2021

Teaching

Instructor, Georgia Tech
ISyE 3103 Supply Chain Modeling: Logistics (undergraduate course) Summer 2026
Responsible for lectures, homework, office hours, and examinations.
Teaching evaluation (4.7/5.0)

Teaching Assistant, Georgia Tech
ISyE 7682 Convexity (graduate course) Spring 2026
ISyE 8803 Pricing and Revenue Management (graduate course) Spring 2023

Guest Lecturer, Georgia Tech
ISyE 8803 Optimization and Learning in Operations (graduate course) Spring 2025
Delivered four lectures on policy gradient methods and nonconvex optimization.

Conference Presentations

Benign Nonconvex Landscape for Policy Optimization: Infinite-Horizon Discounted MDPs with General State and Action Spaces.

INFORMS Annual Meeting, San Francisco, CA Nov 2026
(Poster) Cornell Young Researchers Workshop, Ithaca, NY Oct 2026

Hidden Convexity in Queueing Models.

INFORMS Annual Meeting, San Francisco, CA Nov 2026
Analytics for X Conference, Singapore Sep 2026
INFORMS RMP Conference, Ann Arbor, MI Jul 2026
INFORMS IOS Conference, Atlanta, GA Mar 2026
(Poster) Cornell Young Researchers Workshop, Ithaca, NY Oct 2025

A Simple Solution to Improving Human Supervision of Algorithms: Evidence from Smart Vending.

INFORMS Annual Meeting, San Francisco, CA — Nov 2026
Annual POMS Conference, Reno, NV — May 2026
INFORMS MSOM Conference, Supply Chain Management SIG, London, UK — Jun 2025
Conference on Digital Experimentation @ MIT, Boston, MA — Oct 2024

Landscape of Policy Optimization for Finite Horizon MDPs with General State and Action.

INFORMS MSOM Conference, Boston, MA — Jul 2026
INFORMS IOS Conference, Atlanta, GA — Mar 2026
(Poster) NeurIPS 2025 ML×OR Workshop, San Diego, CA — Dec 2025
INFORMS Annual Meeting, Atlanta, GA — Oct 2025
International Conference on Continuous Optimization, Los Angeles, CA — Jul 2025
INFORMS RMP Conference, New York, NY — Jul 2025
INFORMS APS Conference, Atlanta, GA — Jun 2025
INFORMS Annual Meeting, Seattle, WA — Oct 2024
(Poster) Cornell Young Researchers Workshop, Ithaca, NY — Oct 2024
Purdue Operations Conference, West Lafayette, IN — Aug 2024
INFORMS Annual Meeting, Phoenix, AZ — Oct 2023

Service

Session Organization

INFORMS Annual Meeting — Nov 2026
Session: Inventory Management in Logistics: Policies, Learning, and Asymptotic Insights

International Conference on Continuous Optimization (ICCOPT 2025) — Jul 2025
Session: Recent Advances in Optimization for MDPs and Stochastic Optimal Control

Reviewer

Operations Research, *Management Science*, *Manufacturing & Service Operations Management*, *Naval Research Logistics*, *Computational Optimization and Applications*

References

Xin Chen (Charles and Nancy Davidson Distinguished Chair Professor) xin.chen@utdallas.edu
Naveen Jindal School of Management, The University of Texas at Dallas | *Ph.D. advisor*

Linwei Xin (Associate Professor) lx267@cornell.edu
School of Operations Research and Information Engineering, Cornell University | *Research collaborator*

Brian Rongqing Han (Assistant Professor) brianhan@illinois.edu
Gies College of Business, University of Illinois Urbana-Champaign | *Research collaborator*

Yifan Hu (Assistant Professor) yifan.hu@rutgers.edu
Department of Statistics, Rutgers University | *Research collaborator*

Guanghui (George) Lan (A. Russell Chandler III Chair and Professor) george.lan@isye.gatech.edu
H. Milton Stewart School of Industrial and Systems Engineering, Georgia Institute of Technology
Dissertation committee member